

Vectors, Dimensions, & Metrics

UChicago Political Science Math Camp

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A Foundational Social Science Question

Question

How do we describe things?

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How do we describe things?

A bit more formally...

How do we encode information in order to analyze phenomena?

A Look Back

We use **sets** to describe certain information about characteristics, attributes, or measurements.

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- ▶ The reals: VDEM index, HDI
- ▶ The rationals: population density, GDP per capita (with a twist – why does this work?)
- ▶ The naturals: vote totals, legislature size, regime length
- ▶ The integers: changes in seat totals, ordinal preference rankings about a median

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How might we define a set for each of these examples?

A degenerate example: the set of recorded entries.

Question: Is this useful to define ex ante?

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 - ▶ An n -dimensional *vector* encodes n different characteristics.
- ▶ Think: one dimension, “2D plane,” “three-space”, etc.

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- ▶ Think: one dimension, “2D plane,” “three-space”, etc.

An important note: Don't get too bogged down in the terminology here. The key concept is that we are encoding multiple measurements into a single object.

One Dimension



One Dimension

- ▶ Our standard framework, measuring one characteristic at a time.
- ▶ Can use our fundamental sets ($\mathbb{R}, \mathbb{Q}, \mathbb{N}, \mathbb{Z}$), or any other sets of our choosing

Qualitative

The appropriate discrete set

Quantitative

The number line

Question: why can't we (always) use a number line analogue for qualitative characteristics?

The Good Stuff

To describe the *possible set* of multiple characteristics, we take the outer product of the relevant dimensions:

$$\begin{aligned} & \{\text{Char. 1}\} \times \{\text{Char. 2}\} \times \dots \times \{\text{Char. } n\} \\ &= \prod_{i=1}^n \{\text{Char. } i\} \end{aligned}$$

This means that all observations will be vectors of the form:

$$(\text{Char. 1}, \text{Char. 2}, \dots, \text{Char. } n)$$

Example: Location

We use latitude and longitude to describe geographic location:

$$\begin{aligned}\text{Earth} &:= \{\textit{Longitude}\} \times \{\textit{Latitude}\} \\ &:= \{x \in \mathbb{R} : -90 \leq x \leq 90\} \times \{y \in \mathbb{R} : |y| \leq 90\}\end{aligned}$$

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We live in 3-space. What assumption makes a 2-space measurement sufficient?

What Comes Next?

You usually can't visualize beyond three-space. In the rare event that you can, you usually shouldn't. But, you can express $\mathbb{R}^4$, $\mathbb{R}^5$,...

Beyond an object's position, we could consider density, temperature, color, favorite album...

Most (all?) quantitative analyses can't use unprocessed qualitative data, so we transform qualitative observations into quantitative variables.

A Not-So-Scary Formulation

Real Space

Suppose we want to identify n characteristics. Then, we generally use $\mathbb{R} \times \mathbb{R} \times \dots \times \mathbb{R}$. This is our construction of $\mathbb{R}^n$: “real space.”

Vectors

A **vector** is an element of $\mathbb{R}^n$ (at least in this characterization).

A Brief Question

Why use $\mathbb{R}^n$ so generally? Why not $\mathbb{N}^n$ for entirely discrete data, or $\mathbb{R}^\ell \times \mathbb{N}^k$ for a combination of ℓ continuous values and k discrete ones?

Measuring Vectors

Suppose $\mathbf{x} \in \mathbb{R}^n$ is a vector such that $\mathbf{x} = (x_1, x_2, \dots, x_n)$.

We can characterize the *length* or *norm* of $\mathbf{x}$ as its distance from $\mathbf{0}$:

$$\|\mathbf{x}\| = \left(x_1^2 + x_2^2 + \dots + x_n^2\right)^{1/2}$$

(How is this connected to our absolute value operator?)

Comparison

We often (always?) want to compare multiple elements, vectors, observations, etc.

But, how we make these comparisons is non-trivial.

An Example of Non-Triviality

Suppose we take all 200+ countries in the world, and we compare each possible pairing (200^2 tuples). Which two countries are the most (or least) similar?

- ▶ By distance from center-to-center
- ▶ By most similar climates
- ▶ By most similar population/ethnic consistency

Are any of these right? Are any even wrong?

A **metric** is a function that returns the distance between two elements in a set. We commonly refer to a metric as “ d ” such that $d(a, b) \in \mathbb{R}_+$

Though we can choose metrics rather freely, there are three conditions every metric must satisfy.

Triangle Inequality

The Triangle Inequality is a key foundation of algebra, and states that for any elements x, y, z :

$$d(x, z) \leq d(x, y) + d(y, z)$$

That is – no path between any two elements can be shorter than the “direct” route.

Triangle Violations

This seems rather obvious, but we can break it relatively easily: Suppose

$d(x, y) = (x - y)^2$. Does this hold for $(x, y, z) = (0, 1, 2)$?

Symmetry

Symmetry is perhaps our simplest condition: distance is equivalent in both directions.

A metric d is symmetric iff $d(x, y) = d(y, x)$ for all x, y .

Suppose $d(x, y) = x - y$. Then, $d(2, 1) = 1$, but $d(1, 2) = -1$. So, symmetry is violated.

Positive Definiteness

A metric d is **positive definite** iff it satisfies two conditions:

1. For all x, y , we have $d(x, y) \geq 0$;
2. $d(x, y) = 0 \iff x = y$.

That is: no negative distances, and non-equivalency must return non-zero distance.

The Discrete Metric

A very simple metric we often consider is called the **discrete metric**, which returns a binary equivalency value:

$$d(x, y) = \begin{cases} 1, & x \neq y \\ 0, & x = y \end{cases}$$

Is this a valid metric? Is it useful in the social sciences?

More Realistically...

We have a standard “default” metric that we call **euclidean distance**:

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \dots + (x_n - y_n)^2} \\ &= \left(\left(\sum_{i=1}^n |x_i - y_i|^2 \right) \right)^{1/2} \end{aligned}$$

A Generalized Approach

Euclidean distance is one iteration of what we call the “ p -metric”:

$$d_p(\mathbf{x}, \mathbf{y}) = \left(\left(\sum_{i=1}^n |x_i - y_i|^p \right) \right)^{1/p}$$

In the social sciences, if we want to use $p \neq 2$, we generally need to justify why...